

stress-test the method’s ability to handle parameter competition and mitigate negative transfer from cross-task learning.

The results presented in Table 13 highlight a critical challenge in in-domain merging. Most baseline methods exhibit a significant performance degradation in the presence of the distractor tasks, a phenomenon akin to catastrophic forgetting. This decline can be largely attributed to the interference from heterogeneous task types, conflicting entity label sets, and the semantic divergence introduced by unrelated domains. In stark contrast, our Evo-Merging approach demonstrates remarkable resilience. By leveraging our specialized denoising and sign-flip scaling mechanisms, Evo-Merging effectively mitigates the negative influence from these unrelated tasks. It successfully filters parameter-level noise and harmonizes conflicting parameter directions, allowing it to distill shared, underlying knowledge from the diverse model pool. Consequently, rather than suffering from interference, our method thrives in scenarios with a greater number of models, achieving superior performance by leveraging the increased diversity.

J Full Per-Task Result Tables

J.1 Detailed Per-Task Results (Llama 3.1)

OOD Full Results (Llama 3.1).

ID Full Results (Llama 3.1).

J.2 Ablation Full Results

Table 14: Full per-task results of our methods and the selected baselines over out-of-domain (OOD) tasks. Evo-Merging and LoRAHub operate on the full repository; other baselines follow the pre-selection strategy described in Appendix G.

Method	NER				RE				ET				Avg	
	Mit-Restaurant		FindVehicle		New York Time		Conll04		Mit-Movie		FabNER			
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1
TIES	21.40	24.22	29.87	29.45	12.55	7.57	20.78	19.05	48.67	51.12	15.91	11.42	24.86	23.81
DARE (TIES+)	10.88	14.31	24.13	23.59	3.69	2.53	10.85	8.08	44.57	48.60	15.44	10.71	18.26	17.97
Task Arithmetic	12.02	11.57	14.29	14.96	15.24	16.04	11.44	12.03	36.70	41.36	9.90	9.68	16.60	17.61
Breadcrumbs	38.99	34.06	47.41	38.42	34.05	28.03	26.30	24.56	49.73	51.60	20.53	16.84	36.17	32.25
Della	29.69	26.06	16.66	17.04	3.94	3.25	19.91	17.97	43.85	47.68	14.41	12.51	21.41	20.75
EMR-Merging	31.89	27.46	46.92	42.25	32.49	28.94	16.45	17.10	46.57	47.35	19.26	17.86	32.26	30.16
LoRAHub	65.05	63.08	31.81	30.75	54.23	48.61	21.30	21.77	41.99	41.70	41.94	39.74	42.72	40.94
Best Adapter	37.71	35.99	39.11	38.49	45.55	38.77	11.47	11.33	46.76	46.16	48.45	47.16	38.18	36.32
LoRA	28.18	29.65	52.19	52.34	52.05	44.66	45.42	41.60	25.20	25.38	18.74	20.01	36.96	35.61
(IA) ³	36.18	41.64	24.98	26.05	30.54	30.70	17.01	19.97	38.80	42.17	15.38	16.70	27.15	29.54
TSV-M	27.88	26.04	7.71	3.38	55.26	47.28	16.67	15.39	76.75	76.65	0.61	0.53	30.81	28.21
AdaMerging	12.70	10.84	42.43	30.52	48.17	40.64	9.39	9.62	78.10	78.75	24.45	23.94	35.87	32.39
Robust Merging	27.63	27.04	0.08	0.05	38.39	33.39	17.17	17.55	66.72	66.37	1.17	1.07	25.19	24.25
Evo-Merging	66.29	65.99	49.61	47.64	57.81	51.15	41.87	43.00	64.11	62.55	43.10	42.47	53.80	52.13

Table 15: Full per-task results of our methods and selected baselines over in-domain (ID) tasks.

Method	ACE_2005		CrossNER_AI		CrossNER_Lit		FabNER		Mit-Movie		MultiNERD		TweetNER		WikiANN		Avg	
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1		
Individual model	70.45	69.37	9.51	21.23	20.05	11.83	48.61	47.09	74.27	74.38	86.15	85.55	62.24	58.26	81.33	80.82	56.58	56.07
TIES	16.16	13.11	18.99	18.93	27.27	26.47	21.02	18.91	54.77	51.24	40.70	48.99	34.05	33.96	55.30	56.47	33.53	33.51
DARE	12.72	13.50	20.40	18.42	25.97	25.13	19.11	18.47	53.12	51.67	51.58	47.14	32.97	32.91	54.13	55.58	33.75	32.85
DELLA	15.73	12.74	17.72	18.02	25.52	25.16	19.98	18.55	54.65	51.94	38.69	47.11	31.99	32.30	54.61	56.15	32.36	32.75
Breadcrumbs	19.74	15.34	21.33	20.03	31.87	29.00	21.18	17.43	51.46	46.91	48.44	55.53	37.26	33.65	56.56	56.82	35.98	34.34
Task Arithmetic	17.37	13.40	20.12	18.90	30.55	28.38	20.86	17.63	53.59	49.27	45.43	53.04	35.60	33.47	56.33	56.93	34.98	33.88
LoRAHub	28.36	25.18	33.67	33.69	33.23	31.51	26.04	22.32	38.90	36.48	36.61	43.89	26.99	30.04	53.78	55.32	34.70	34.80
EMR-Merging	19.08	15.10	16.99	16.16	26.09	23.12	23.70	19.76	52.09	46.67	46.65	54.16	37.91	34.76	59.01	59.49	35.19	33.65
TSV-M	43.03	37.32	27.57	25.41	21.40	18.90	31.03	27.29	52.85	49.76	30.19	36.37	31.78	33.47	55.09	55.54	36.62	35.51
AdaMerging	13.37	8.19	38.30	29.00	27.80	23.24	24.77	12.88	34.90	26.54	74.53	74.88	31.60	26.73	59.44	57.97	38.09	32.43
Robust Merging	20.72	15.73	37.60	34.93	32.80	30.18	24.78	18.92	46.28	40.51	45.05	52.06	40.01	40.10	61.76	61.57	38.63	36.75
Evo-Merging	26.31	23.46	34.18	34.61	33.95	32.32	27.15	23.27	43.58	39.94	52.82	56.59	36.34	35.08	58.40	58.98	39.09	38.03

Table 16: Full per-task ablation study of Evo-Merging's components. [↓] means the performance drop relative to our full model.

Method	NER_Mit-Rest		NER_FindVeh		RE_NYT		RE_Conll04		ET_Mit-Movie		ET_FabNER		Avg	
	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1	Prec	F1		
Evo-Merging	66.29	65.99	49.61	47.64	57.81	51.15	41.87	43.00	64.11	62.55	43.10	42.47	53.80	52.13
w/o Denoising & Scaling	28.01	33.10	44.14	32.68	2.35	1.94	7.36	5.37	43.70	39.67	14.10	11.75	23.28 [↓ 30.5]	20.75 [↓ 31.4]
w/o Denoising	65.05	63.08	31.81	30.75	54.23	48.61	21.30	21.77	41.99	41.70	41.94	39.74	42.72 [↓ 11.1]	40.94 [↓ 11.2]
w/o Scaling	39.90	37.41	48.78	41.35	2.27	2.23	7.14	6.38	46.10	46.52	14.10	11.75	26.38 [↓ 27.4]	24.27 [↓ 27.9]
w/o Sign-flipped	42.04	37.70	46.61	41.07	45.45	38.38	35.21	32.91	62.19	60.47	34.77	29.80	44.38 [↓ 9.4]	40.06 [↓ 12.1]

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